

Fall 2024 Project Report



Identifying travel needs, barriers, and solutions for transport-disadvantaged populations

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1. Introduction

Transportation plays a critical role in our daily lives, providing access to essential services, employment, education, healthcare, and social activities. This access shapes life opportunities and affects our overall sense of possibility. The degree to which individual needs are satisfied through out-of-home activities is determined by the dynamic interplay among individual preferences, desires, resources (e.g., mode availability and level of service), and abilities (e.g., health status), skills, and other contextual conditions for mobility (e.g., urban form) (Nordbakke & Schwanen, 2015; Tonn, 1984). The current transportation system evidences disparities in the distribution of benefits and burdens for different population groups (Karner et al., 2016). These disparities are not randomly distributed; rather, they are patterned upon existing systems of advantage and disadvantage based on race/ethnicity, income, gender, age, disability, and other social identities. Individuals who belong to socially dominant groups, such as those who are white, wealthy, or male, are often positioned as advantaged. Conversely, those who belong to marginalized groups, such as people of color, low-income individuals, or women, are often disadvantaged. Such systemic inequality means that certain groups consistently have greater access to resources and opportunities, while others face structural barriers that limit their mobility, accessibility, and overall life satisfaction.

Transport equity refers to ensuring fair and equitable access to transportation resources for all individuals, regardless of their social identities, with the imperative to address those who are least advantaged (Karner et al., 2020; Lucas et al., 2016). Groups facing significant transportation barriers include those with no or limited car access, low-income individuals, women, older adults, single parents, Black, Indigenous, and people of color (BIPOC), immigrants and non-native residents, youth, people with disabilities, the unemployed, rural populations, shift workers, parents traveling with children, and students (Currie & Delbosc, 2011). These socially disadvantaged groups face mobility and accessibility-related challenges in their daily travel. Accessibility poverty refers to a state where an individual's ability to reach essential activities like work, school, healthcare, and leisure becomes so low that fulfilling basic needs becomes impossible (Karner et al., 2023; Martens & Bastiaanssen, 2019). Identifying the travel needs and barriers that lead to accessibility poverty for transportation-disadvantaged groups is a crucial step towards creating an equitable and just transportation system.

In the U.S. context, several nationally representative data sources can be used to identify travel needs and barriers. These include the National Household Travel Survey (NHTS), the American Time Use Survey (ATUS), and the American Community Survey (ACS). The NHTS is administered by the Federal Highway Administration (FHWA). Results from the survey can provide insights into the transportation challenges faced by different groups—they allow researchers to analyze how factors such as race, ethnicity, age, immigration status, disability status, and socioeconomic background affect travel patterns. The ATUS, conducted by the U.S. Bureau of Labor Statistics, measures the amount of time people spend on various activities, including commuting and other travel-related tasks. The ATUS is essential for understanding the time-use patterns and travel behavior of individuals, offering insights into how time constraints and daily routines impact transportation needs and choices. The ACS, administered by the U.S. Census Bureau, provides annual data on commute patterns, modes of transportation,

employment, income, and housing, giving a broad view of the socioeconomic factors influencing travel behavior.

One primary difficulty in understanding travel needs, especially those that are unmet, lies in the inherent limitations of traditional travel surveys, such as the NHTS, which focus on capturing observable and tangible aspects of travel behavior—namely, the number of trips, distances traveled, and modes used, among others. While these surveys excel at detailing what trips are taken and for what purpose, they fall short in identifying the trips that are desired but not taken due to various barriers. Unmet travel needs often stem from factors that are not easily quantifiable or directly observable. Additionally, surveys typically quantify travel behavior but do not delve deeply into the subjective experiences of travelers, such as their frustration with unreliable public transport or the stress of navigating unsafe walking paths, satisfaction and perceived accessibility. These qualitative aspects of travel are essential for understanding the barriers that lead to unmet travel needs but are often neglected in traditional surveys.

The primary objective of this report is to analyze the key findings and gaps in past research that have used the NHTS to understand the travel needs and barriers discussed for different population groups. We reviewed the questionnaires used for the past four NHTS iterations conducted in 2001, 2009, 2017, and 2022 to understand how different questions and their options can contribute to our understanding of various travel needs, barriers, and overall transportation equity issues. Our goal is to identify gaps in the current NHTS questionnaires and suggest additional questions that could be included in future surveys. These questions should be specifically designed to capture the unique experiences and challenges faced by transport-disadvantaged populations, including low-income individuals, BIPOC populations, older adults, and people with disabilities. The findings reported upon here will help ensure that future surveys are more inclusive and effective in capturing the nuanced transportation experiences of all population groups.

2. Topics and questions relevant to transportation equity within the NHTS

The NHTS collects nationally representative data on demographics, household characteristics, vehicles, and multimodal statistics on personal travel. From these data, several key travel metrics that help to understand mobility patterns can be calculated. These include, but are not limited to: trips per person, person-miles traveled (PMT) per person, trip counts by purpose, vehicle trips per driver, vehicle-miles traveled (VMT) per driver, average vehicle trip length, and average time spent driving. The importance of these statistics lies in their ability to provide a comprehensive overview of travel behavior across different population groups. For instance, person-trips per person and PMT per person help to quantify the total travel undertaken by individuals, reflecting overall mobility. Vehicle trips per driver and VMT per driver provide insights into the dependency on personal vehicles and the extent of road usage. Average vehicle trip length and average time spent driving offer detailed information on trip characteristics, helping to identify trends in travel efficiency and congestion levels. However, these metrics also have limitations when it comes to studying transportation needs and barriers. The next section discusses specific questions from the 2001-2022 NHTS iterations, detailing how they contribute to our understanding of travel needs and barriers. Specific variables and their description are listed in Appendix A.

2.1 Unmet need or latent demand

Addressing transportation equity necessitates a comprehensive understanding of not just trips taken, but also those that are missed or forgone due to various barriers. Latent demand refers to the activities and travel that individuals desire but are unable to realize due to various constraints, such as inadequate transportation infrastructure, financial limitations, or lack of awareness (Clifton & Moura, 2017). The NHTS has attempted to capture the concept of latent demand in the past. For example, the 2022 NHTS asked respondents whether they have made fewer trips in past 30 days along with reasons behind it. The response set included options such as “didn’t feel safe,” “didn’t feel clean/healthy,” “not reliable,” “didn’t go where needed,” and “unaffordable.” These possible responses potentially indicate barriers or constraints related to travel options and travel quality. Other options that respondents could select included health problems, not enough time, COVID-19, and online deliveries, among others.

The 2017 and 2022 iterations of the NHTS offered specific reasons for those who did not make any trips on travel day. Earlier versions of the NHTS only asked respondents whether they stayed in the same place all day. The percentage of responses indicating no travel helps identify the level of immobility within the population. Respondents to the 2017 and 2022 NHTS were able to select different reasons for not traveling that encompass a range of personal, employment, and environmental factors. Personal reasons included being sick or quarantining, hospitalized, on vacation or taking a personal day, being reluctant to travel, or being out of the country. Employment conditions, such as working at home (for pay) and not being scheduled to work, could also be selected. Environmental factors like bad weather were included, and other reasons such as household chores/projects and caretaking duties were also included. Several options were directly related to travel problems. These included a lack of available transportation, indicating forgone trips due to transportation-related issues. For example, certain groups may be more likely to report no travel due to the unavailability of transportation options, which reveals patterns of immobility tied to systemic inequities.

In the 2001, 2009, and 2017 NHTS, non-traveling respondents were additionally asked about the approximate number of days since their last trip before the travel day. In the 2017 survey, the options were categorized as follows: the day before, a few days before, a week before, more than a week before but within a month, more than a month before. In contrast, the 2001 and 2009 surveys did not use categorical options; instead, respondents were asked to directly input the number of days since their last trip. The NHTS editions of 2001 and 2009 included a question about whether respondents wanted to get out more often, which provided insights into unrealized travel demand. This question would help to identify individuals who had a desire to travel but faced barriers that prevented them from doing so. Unfortunately, this question was not continued in subsequent NHTS iterations, resulting in a gap in understanding unrealized travel demand in more recent datasets.

2.2 Affordability

The 2017 NHTS included questions that directly addressed transportation affordability. Respondents were asked whether travel constitutes a financial burden for their household and whether the price of gasoline affects their travel decisions. These questions aimed to gauge the extent to which financial constraints influence travel behavior and decisions. The survey also inquired about how people adapt their travel behavior in response to financial burdens. Options

provided included whether they use alternative travel modes—walking, biking, or public transit—to reduce travel costs. These responses help in understanding the strategies individuals and households employ to cope with financial constraints related to transportation. For example, respondents who report shifting to walking, biking, or public transit due to financial constraints indicates that affordability is a likely barrier that impacts their mobility. The inclusion of the question "Price of gasoline affects travel" in the 2017 NHTS contributes to the affordability dimension by quantifying the extent to which fuel costs influence individual travel behaviors, providing valuable insights into economic barriers to mobility. Furthermore, the survey introduced a question regarding the use of discounted transit passes in the past 30 days. Analyzing these responses can reveal the reach and effectiveness of discounted transit pass programs, particularly for low-income groups, students, and older adults. By analyzing these questions, researchers can identify which population groups are most affected by transportation-related financial burdens. Another useful addition in the 2022 NHTS is the detailed information it gathers on parking at the household and trip levels. The survey asked respondents about the amount paid for parking, and the periodicity of these payments, such as one-time or monthly payments. These details help elucidate the financial burden of parking on households, revealing potential disparities based on location and socioeconomic status. Additionally, questions distinguishing between self-paid parking and parking costs covered by employers or educational institutions highlight inequities in transportation benefits and support structures.

2.4 Perceived broad transport barriers

In the 2001 and 2009 NHTS iterations, respondents were asked to express their views on what they considered to be the most critical transportation issues. The survey provided options encompassing safety, highway congestion, travel cost, issues with aggressive or distracted drivers, the availability and accessibility of public transit, and the absence of adequate walkways or sidewalks. This question is particularly useful for understanding perceived transportation problems from a diverse population, giving a voice to a wide array of travelers, and revealing common frustrations and barriers across different demographics. Moreover, insights derived from such questions can help identify geographic and demographic disparities in transportation needs. For instance, if residents in certain areas report high levels of dissatisfaction with public transit access, it may indicate a need for expanding services in those regions. Similarly, if certain groups, like older adults or people with disabilities, consistently report specific barriers, it can lead to tailored solutions that address their unique needs. However, this question was removed in later NHTS iterations, limiting the ability to analyze trends over time and track changes in public perception and the impact of past transportation policies.

2.5 Impact of COVID-19

The 2022 NHTS placed significant focus on understanding how the COVID-19 pandemic affected travel behavior. Specifically, the NHTS 2022 asked respondents: “in the last 30 days, did this person take fewer trips than they had planned on taking for any reason?” The response set included options directly related to the transport system, such as affordability, reliability, coverage, safety, and cleanliness/health concerns. Additionally, the survey captured personal factors, including time availability, health issues, and the increase in online deliveries due to the pandemic. Another key question asked respondents how their current travel and activities compare to their travel before the pandemic. Complementary questions addressed changes in work and physical school/class locations, online shopping habits, and public transit use, with

respondents indicating whether these changes are temporary or permanent. Other questions sought to capture the nature of online deliveries at the person level.

Examining how pandemic-induced changes to schedules and routes may have had different effects across people and places is a critical research question. The NHTS questions related to changes in public transit use can shed some light on the effects of service changes and their distribution across the population, for example by examining which demographic groups have experienced reduced travel compared to pre-pandemic levels, which groups have increased their travel, and which have remained the same.

2.6 First mile-last mile connection

The concept of the "first mile-last mile" connection is pivotal for understanding and improving the accessibility and efficiency of public transit systems. This connection refers to how travelers get from their starting location to a transportation service (first mile) and then from the service to their destination (last mile). The 2001, 2009, 2017 NHTS included information on the modes used to access public transit, with options including bus, rail, personal vehicle (POV), subway, walking, and other modes. The survey aimed to determine if respondents used these modes as part of their public transit journey, providing insight into the commonality and efficiency of various first and last mile solutions. The 2017 survey also included questions about the time taken to reach transit stations and the time from stations to the final destination. These questions offered a range of responses from 0-300 minutes, highlighting the time commitment involved in transit-linked travel.

2.6 Mode choice

The NHTS in general provides a wealth of derived variables that are instrumental in understanding key travel outcomes like mode choice. Respondents were asked about their frequency of using various modes, including walking, bicycling, bikeshare, e-scooters, motorcycles, public transit, rideshare, and taxi services. The 2017 and 2022 NHTS specifically included questions on the use of emerging travel modes like rideshare, bikeshare, and e-scooters. Respondents also reported using these modes for travel to work, school, and on the actual travel day. These are useful for understanding the adoption of new transportation technologies and their impact on travel behavior.

The 2017 NHTS asked respondents about why they do not walk or bike more often and the 2009 survey asked about general issues related to walking and biking. Possible responses included a detailed list of potential barriers such as being too busy, experiencing poor health, having a lack of companions, the absence of nearby paths or trails, insufficient bike or wide curb lanes, poor sidewalk conditions, unsafe street crossings, lack of interesting destinations, fear of street crime, too many cars, fast traffic, air pollution, carrying heavy loads, accompanying small children, and inadequate lighting at night. Understanding these barriers can inform interventions aimed at increasing active transportation. The 2017 survey did not include specific questions about school travel, unlike the 2009 survey, which addressed safety concerns related to walking and biking to school. The 2009 survey asked respondents to agree or disagree with statements about potential barriers such as violence/crime along the route, distance between home and school, traffic speed, and the volume of traffic. This data helps in understanding parental concerns and the factors influencing children's mode choice for school travel.

The 2017 and 2022 NHTS explored mobility challenges faced by individuals who rely on various medical devices. The survey includes a comprehensive list of medical devices that assist in mobility (e.g., Cane/Walking stick, motorized scooter or wheelchair, devices to aid the blind or visually impaired, walker or crutches, dog assistance). Understanding the types of medical devices used by travelers can help transportation planners and service providers tailor their services to better accommodate these needs. For instance, transit authorities might use this data to enhance vehicle accessibility, improve station design, or adjust policies around service animals and mobility aids.

2.7 Disadvantaged people/communities

The NHTS collects a wide range of demographic data that are critical for identifying different population groups that face social and transport disadvantage. These data include age, gender, race and ethnicity, education level, income, employment status, physical condition, car ownership, internet access, immigrant status, and language proficiency, among others. For instance, information gathered about car ownership and whether respondents personally drive provides valuable insights into households that lack access to a private vehicle. These households may encounter significant transportation barriers, affecting their ability to access essential services and participate fully in daily activities. Demographic data collected at the household level include household size and composition, vehicle availability, household income, home ownership, life cycle classification, and location. Data on household size, the presence of children, and single-parent households help understand the specific transport needs of larger or single-parent families. Information on the number of vehicles per household, drivers per household, and vehicles per driver/workers per household sheds light on households with limited vehicle access. Household income levels and home ownership status provide context for understanding financial constraints and housing stability, which can impact transport access.

Over the years, from 2001 to 2022, the NHTS has progressively enhanced understanding of how health conditions and disabilities influence travel behaviors. The survey collects data on a range of travel modifications and adaptations that people make due to their health conditions or disabilities. For example, respondents have indicated changes such as limited driving during daytime, using buses or subways less frequently, relying on others for rides, or entirely giving up driving. Additionally, some have turned to alternatives like rideshare or specialized transportation services tailored for people with disabilities, and these adaptations reflect significant shifts in their mobility patterns. Responses to these survey items shed light on the substantial impact that health conditions can have on one's ability to travel independently. For instance, the selection of “Given up driving” or “Used special transportation services” not only highlights personal adjustments but also underscores the broader demand for accessible transportation options. The inclusion of such detailed inquiries helps policymakers understand the extent of travel modifications necessitated by health issues, guiding improvements in transportation planning to accommodate these specific needs.

2.8 Built environment

NHTS includes detailed built environment data on respondents' home, origin, and destination trip locations through tract and block group variables. These variables include housing units per square mile, percent renter-occupied housing, population density, and an urban/rural code which

categorizes locations into five types: urban, rural, suburban, second city, and small town. These classifications are critical as they help distinguish between different types of environments, providing nuanced insights into how various neighborhood contexts influence travel behavior. Furthermore, the NHTS consistently provides information on the household's location within census divisions and regions, the population size of the relevant metropolitan statistical area (MSA), urban area classifications, urban area size, and the presence of rail services. These geographic variables are vital for analyzing travel behavior patterns as they offer a comprehensive perspective on the environmental and socio-economic contexts in which households operate. For instance, understanding whether a household is in a high-density urban area, or a low-density rural setting can reveal significant differences in access to transportation options and services, influencing mobility and accessibility outcomes. In the 2001 and 2009 NHTS, respondents were asked to identify the most important reasons for staying in their current home and choosing their current home location. This information helps understand the underlying factors influencing residential stability and locational choices.

3. Studying transport-disadvantaged populations using the NHTS

In this section, we summarize findings from studies that have used the NHTS, or earlier versions of the survey known as the National Passenger Travel Survey (NPTS), as their primary data source to understand the mobility-related challenges faced by various disadvantaged groups. We assumed that studies focusing solely on mobility outcomes without a disaggregated analysis of socio-demographics or social groups offer limited insight into transport equity issues and policy implications. Our review covers studies published between 1995 and 2024 that used the NHTS data series, sourced from the Web of Science database. To ensure a comprehensive overview of mobility, accessibility, and travel behavior research, we employed a robust set of keywords encompassing various transportation modes and equity-focused terms.

3.1 Older adults

Prior studies have illustrated the mobility-related challenges faced by older adults, including engaging in fewer trips, covering shorter travel times, and traveling shorter distances (Collia et al., 2003; Wasfi et al., 2012). Although shorter travel times and distances are generally seen as desirable, for this population group, they might indicate barriers that limit their ability to reach suitable or desirable destinations, which might otherwise involve covering more distance or time. The integration of new transport modes such as automated vehicles could redefine travel patterns for older adults, potentially increasing their vehicle miles traveled (VMT), as described by Harper et al. (2016). Technological adaptations, however, are not uniformly accessible. Mitra et al. (2021) highlight significant mobility gaps following driving cessation, particularly among older women of color, a group further marginalized by socio-economic factors like income and education levels. Yet, there is a silver lining as more affluent and educated older adults increasingly adopt ride-hailing services, illustrating a shift towards flexible, on-demand transport solutions (S. K. Mitra et al., 2019).

3.2 People with disabilities

Several studies have leveraged the NHTS to examine transportation patterns and disparities among people with disabilities in the United States. Henly & Brucker (2019) found that individuals with disabilities were less likely to travel for errands, shopping, social, or recreational

reasons, especially those with lifelong disabilities. Cochran & Chatman (2021) revealed that adults with disabilities use app-based ride-hailing services like Uber and Lyft less frequently than those without disabilities, even after controlling for factors like age, income, and urban residence, indicating physical accessibility issues. Eisenberg et al. (2022) demonstrated that while people with disabilities are less likely to use rideshare services, those who do adopt them use these services more frequently than non-disabled individuals, with usage patterns influenced by health, age, race, education, employment, and residential density. Myers et al. (2022) highlighted that disabled adults in rural areas face greater transportation challenges compared to their urban counterparts, relying more on personal vehicles and less on public transit or walking. These findings collectively underscore the significant transportation barriers faced by disabled individuals and the need for improved, accessible transportation options to enhance their community participation and mobility.

3.2 Low-income people

Several studies have used data from the NHTS to investigate the travel behaviors of low-income households in the United States. Blumenberg & Pierce (2012) examined the 2009 data, revealing that low-income adults quickly convert rising income into increased mobility, particularly benefiting from the transition from zero to one vehicle, which significantly enhances their travel capabilities. In another study, Blumenberg & Pierce (2014) found that low-income individuals engage in less multimodal travel compared to higher-income individuals, but when they do, they take more trips, suggesting the need for better multimodal travel options to enhance mobility for the poor. Blumenberg et al. (2020) elucidated the impact of vehicle ownership status—comparing zero-car households, car-deficit households, and fully equipped households—on various travel outcomes, including person-miles traveled (PMT), vehicle-miles traveled (VMT), number of trips defining activity participation, and the use of transit as an alternative mode. The study discerned variations in travel patterns among populations residing in diverse residential locations with distinct racial, income, and household compositions. The findings indicate that carless households and car-deficit households (with less than one car per driver) exhibit significantly reduced travel compared to “fully equipped” households (with one or more car per driver). While diminished travel does not inherently imply transport disadvantage for a household, the study suggests a potential connection between not owning a car, household composition (e.g., households with children, households with disabled members), and residing in specific neighborhood types (e.g., suburban or rural areas with limited alternative transport options), which may intensify the daily travel challenges for household members. The research advocates for future investigations into how car deficiency might influence household residential location choices, employment outcomes, and activity participation.

Liu & Uddin (2023) used the 2022 NHTS to identify mobility gaps between low-income and higher-income households in New York state, finding that low-income households take fewer and shorter-distance trips, but experience longer travel times, emphasizing the need for equitable transportation planning to address these disparities. Similarly, Giuliano (2003) found that low-income Hispanic/Latino and Black households showed a more pronounced reduction in travel distance compared to their middle-income counterparts. Jordan et al. (2019) found younger seniors with medium household income experience increased trip durations, while high-income households have decreased durations. For old seniors, high household income increases trip

durations. These studies collectively underscore the critical role of automobile access and the need for multimodal transportation solutions to improve mobility for low-income households.

3.4 Women

Studies reported that women tend to have shorter travel distances and durations compared to men and rely more on public transportation due to gendered roles that prioritize household maintenance and childcare. Kwon & Akar (2022) noted a slight narrowing of the gender gap in commute distances within two-earner households using the 2001, 2009, and 2017 NHTS, with women still having shorter commutes. Shorter commutes, often perceived as advantageous, may signify constrained mobility options tied to traditional gender roles and responsibilities, not true travel efficiency or choice. This study reflects persistent gender inequalities in travel behavior, despite some progress. Jin & Yu (2021) and Kwon & Akar (2022) has documented how socio-economic factors and residential environment influence women's likelihood to utilize public transit and the distances they commute. Jin & Yu (2021) showed that women are less likely to use transit under certain conditions, such as higher household income and low residential density. These patterns are not just isolated statistics but are indicative of deeper, systemic issues tied to the Household Responsibility Hypothesis (HRH) (Turner & Niemeier, 1997), which posits that domestic duties, often disproportionately shouldered by women, significantly dictate their mobility patterns. This hypothesis is further supported by Hu's (2023) findings, which emphasize how these responsibilities impact women's time spent outside the home, especially in households with children. These roles not only restrict women's employment opportunities but also impose spatial-temporal constraints on their daily activities, leading to a significant gender gap in activity fixity and leisure time. Moreover, the intersectionality of gender with race and household structure, as explored by Hu (2023), reveals that these factors collectively shape commuting patterns, with significant variations across different demographic groups. Kim & Ulfarsson (2021) focused on older minority women, finding that low income, low education, rural residence, and non-driving status contribute to high levels of immobility. This study underscores the compounded disadvantages faced by older minority women. Safety concerns further exacerbate disparities, with women more likely to avoid activities during certain times, particularly at night, due to fears of victimization and gender-insensitive urban planning. Studies also emphasized on gender roles in driving behaviors and the implications for transport safety and enforcement policies. Robertson et al. (2011) highlighted the narrowing gender gap in DUI arrests, attributing it to changes in women's driving behavior and law enforcement practices. The theme of gender disparities in pedestrian safety is addressed by Zhu et al. (2013), who found that male pedestrians have higher fatality rates per collision than female pedestrians. These disparities could be rooted in behavioral differences between genders. Research indicates that men may engage in riskier behaviors while walking, such as jaywalking or crossing streets away from designated crosswalks, more frequently than women. This riskier pedestrian behavior could lead to more severe accidents when they occur.

3.4 Immigrants

The literature explored how immigrant status, i.e., place of birth/origin's nationality, and tenure in the United States affect mode choice household and external carpools than native-born Americans, with a stronger tendency for household carpools, although this propensity decreases as time in country increases. Immigrants often co-locate in so-called "ethnic enclaves," maximizing social capital and facilitating transportation resource sharing. Over time, as their

financial situation improves and social networks diversify, reliance on carpooling and ethnic enclaves decreases. The conceptual framework presented by Delbosc & Shafi (2023a) highlights how social factors such as social and travel norms, gender roles, and cultural practices, alongside personal factors like past travel behavior, travel skills, demographics, and attitudes from immigrants' countries of origin, influence the travel assimilation process in the destination country. Therefore, more detailed information on these personal and social dimensions would significantly enhance the understanding of travel needs within immigrant communities.

3.5 Racial and ethnic groups

Past studies highlight disparities in transportation access and preferences among different racial and ethnic groups. While studying NPTS 1995, Giuliano (2003) observed significant differences in travel characteristics across racial/ethnic groups e.g., whites had the highest average daily travel distance and travel time, while black people had the lowest. Even after controlling for socio-economic and geographic factors, race and ethnicity still had significant independent effects on travel behavior. Jimenez & Mattingly (2009) have analyzed the travel behavior of three racial-ethnic minority cohorts in Texas using data from the 2001 Texas NHTS: U.S.-born Hispanic or Latino residents, Hispanic or Latino immigrants, and Black people. U.S.-born Hispanic or Latino people were found to closely emulate the travel behavior of white residents, exhibiting similar trip rates for work and family purposes but producing higher rates of nonwork-related trips, particularly to places of worship and school. Hispanic immigrants, on the other hand, had lower vehicle ownership and consequently drove less, relying more on public transit compared to their U.S.-born counterparts. Black respondents had the lowest mean rate of social- and recreational-trip production and experienced the longest commute times among all groups, reflecting their significant reliance on public transit for commuting. The issue of time burden for transit riders is highlighted by several other researchers. Jordan et al. (2019) found that African-American experience a 41.6% increase in travel times relative to non-African-Americans. They explained this increase may be due to numerous factors including access and availability to transportation, having a driver's license, local infrastructure and built environment, income inequalities, and racial disability rate disparities. Probst et al. (2007) noted that Black people spend more time using transit compared to white people, with an increased likelihood of trips lasting 30 minutes or longer. Labban et al. (2023) identified that non-Hispanic Black respondents with lower incomes face a higher travel time burden when using public transportation compared to their white counterparts. McAndrews et al. (2013) showed that American Indian and Black travelers in Wisconsin faced higher transportation injury risks than white travelers. These disparities are linked to broader socio-economic conditions, neighborhood characteristics, and specific transportation needs and preferences of racial and ethnic groups.

4. Assessing travel needs and barriers beyond NHTS

While the NHTS provides valuable insights into general travel behavior and mobility outcomes at a national scale, it is less effective in addressing specific travel problems within individual communities, particularly among transport-disadvantaged populations. This limitation arises because the NHTS is not designed to gather detailed data on the localized travel challenges faced by these groups. The NHTS predominantly adopts a mobility-based assessment approach. Singer and Martens (2023) outline four distinct approaches for evaluating travel problems—mobility measures are one that they include. The second approach encompasses accessibility-based

measurements that have long been recognized as a valuable tool for gauging the relationship between transportation and land use and capturing the underlying drivers of travel demand—namely the need to reach locations separated in space. Compared to mobility measures, accessibility-based approaches provide a more targeted understanding of how accessible essential services and opportunities are for various populations. The third approach, focusing on travel satisfaction, has gained prominence since the early 2000s, reflecting a shift towards evaluating qualitative aspects of transportation experiences, further highlighting the need for comprehensive approaches that go beyond the broad metrics captured by surveys like the NHTS. The fourth approach involves the direct measurement of travel problems, with a focus on transport-disadvantaged groups. This approach provides a targeted examination of the challenges faced by these groups, offering valuable insights for designing interventions and policies that address their unique needs and barriers. This section highlights studies that use direct travel problem measurement from single or multiple disadvantaged groups for understanding unique travel needs and barriers in the United States. The objective of reviewing these studies is to gain additional insights into people's unique travel needs and barriers, which are often not obtainable from broad assessments like the NHTS. The insights gained from these localized assessments are important for informing policy decisions and improving national scale travel surveys.

A variety of research designs have been employed to understand the travel challenges faced by various transport-disadvantaged groups, encompassing qualitative, quantitative, and mixed-method approaches. Qualitative methods include one-to-one or face-to-face semi-structured interviews (Ward & Walsh, 2023) and focus groups (Nahar & Cronley, 2021). Fields et al. (2019) and Fan et al. (2023) have used custom-designed daily transportation diary apps and smartphone-based travel behavior surveys to collect detailed trip data over extended periods, such as 7 or 14 days. Additionally, several studies have adopted mixed-method approaches that combine in-person interviews or focus groups with online surveys to gather comprehensive data, as seen in the works of Fan et al. (2023), Golub et al. (2019), and Shaheen et al. (2022). Recruitment strategies for participants have varied, with researchers employing flyers and posters in local public housing communities, as well as brochures, print media, and digital platforms like social media (He et al., 2022; Lee et al., 2023; Ward & Walsh, 2023). These diverse methodologies have enabled a deeper understanding of the complex mobility issues these groups encounter.

The literature identifies several factors that act as barriers or constraints to daily travel, encompassing transport system-specific elements (such as temporal and spatial coverage, safety, affordability, infrastructure, and service quality), and individual or personal factors (such as socio-demographic identity, car ownership, skills and abilities, motivations). Most of the studies being reviewed reported how problems with transport leads to unmet need, resulting in reduced activity participation or completely forgoing the activity, which ultimately make the issue of social exclusion. Fields et al., (2020) found that instances where the lack of suitable transport options led to limited travel, forgoing trips, missed opportunities, e.g., medical appointment, and reduced community participation. The study document the reliance on others for transportation, the difficulties in arranging rides in last minute, safety concerns about traveling during the day, cost of travel, and the emotional impact, such as feelings of isolation and depression as consequence of travel problem for older adults. Similarly Ward & Walsh (2023) found low income transit dependent population have experienced latent demand that resulted in adaptive

strategies among the respondent who said they have made limited employment, social activities, and not participating in outdoor activities. Fan et al. (2023) find that underserved communities in Minnesota, e.g., Latinx, African American, Hmong, people with disabilities, immigrants, people living with HIV, single mothers, single fathers, transitioning home residents in Fergus Falls, and tribal members of the White Earth Nation experience diverse transportation barriers, revealing unique and shared challenges. People with disabilities face accessibility issues, such as access to vehicle, leading to missed appointments. Immigrant communities struggle with language barriers that confuse transit navigation, e.g., Hmong-speaking participant couldn't understand transit announcements. Single parents report a lack of family-friendly transit options, making it difficult to manage public transport schedules with children's school timings. Tribal members from isolated areas like the White Earth Nation face infrequent bus services, restricting access to employment. People living with HIV express concerns about using public transit during peak hours due to fear of stigma and discrimination. Nahar & Cronley (2021) studied the daily mobility challenges faced by 15 immigrant intimate partner violence (IPV) survivors in North Texas. Their research revealed that reliance on a partner for transportation, controlling behavior of their partners, combined with a lack of independence due to not owning a car, significantly exacerbates the difficulties these survivors experience. Survivors also report feeling unsafe on public transit, with transit inaccessibility due to shelter locations being far from coverage areas contributing to their transport disadvantage. Additionally, the possibility of encountering partners or relatives during transit journeys, along with longer travel times, safety concerns, and reliability issues, further compounds the challenges they face. LGBTQ+ individuals encounter barriers related to societal prejudices and discrimination, affecting their willingness to use certain modes, e.g., public transport (Lubitow et al., 2017; Weintrob et al., 2021). This detailed account underlines the critical need for tailored transportation policies that address the specific mobility challenges of each group, enhancing their ability to participate fully in societal activities.

Further, few studies explored mode specific barriers. Lee et al. (2023) studied mode-specific challenges (e.g., cars, public transit, and paratransit) using journey mapping technique among 20 individuals with disabilities. They considered barriers in each phase of the journey by a specific mode which included barriers before (considering a trip, planning a trip, locating transportation, entering transportation), during (riding transportation), and after (exiting transportation, arriving at destination) stage of a travel. Their study identified barriers included the need for third-party assistance, scheduling problems, long wait times, challenges with vehicle ingress/egress, difficulties in using transportation apps, lack of knowledge regarding services and schedules, and issues related to the accessibility of service locations, including wheelchair securement. Golub et al. (2019) explore the challenges faced by lower-income and racially diverse communities in Portland, Oregon, in accessing smart mobility systems. The study identifies significant barriers that prevent these groups from fully benefiting from advancements in smart mobility. Key obstacles include limited access to necessary financial tools such as bank accounts and credit cards, a reliance on cash transactions for transit, and reduced availability of internet access. Information security concerns and disparities in internet and data usage further hinder the equitable adoption of smart mobility. Shaheen et al. (2022) identified several critical barriers that significantly impact the transportation choices of underserved populations relying on Transportation Network Companies (TNCs) and public transit. Primarily, the high cost of TNCs, exacerbated by surge pricing during peak times, poses a financial challenge that limits their use, particularly for lower-income individuals. While public transit offers a more affordable

alternative, it is often perceived as unreliable due to inconsistent schedules, overcrowding, and cleanliness concerns, leading many users to prefer TNCs despite their higher costs. They also found safety concern as participants reported fears of harassment and dangerous conditions on public transit, which deters usage, especially among vulnerable groups such as women and members of the LGBTQ+ community. For individuals with disabilities, inadequate accessibility in both TNC services and public transportation creates additional obstacles to mobility. He et al. (2022) studied the impact of the COVID-19 pandemic on essential transit riders across the U.S. Their research identified several issues arising from reduced transit services, such as the inability to reach desired destinations, decreased convenience, and affordability concerns. Safety-related anxieties also emerged, including fears of interacting with police, exposure to COVID-19, harassment, physical harm, and interactions with Immigration and Customs Enforcement (ICE). The types of trips affected included those for healthcare, groceries, and childcare. The reductions in transit services disproportionately impacted specific demographic groups, including low-income households, Hispanic or Latinx riders, and female, non-binary, or genderqueer individuals. While wealthier riders could avoid public transit by working from home or utilizing private transportation, essential workers and vulnerable populations faced greater hardships due to reduced services and increased exposure risks.

The aforementioned studies highlight how the travel needs and barriers faced by various underserved and marginalized communities can be directly measured. These studies are ad hoc in that they collected qualitative and/or quantitative data from relatively small samples to understand the transportation disadvantage conditions specific to groups in specific locations. None of the work envisions integrating these methods into large-scale travel behavior surveys, nor do they simultaneously address how to measure and assess transportation disadvantage across a broader population. In next section, we will discuss about our approach to develop a comprehensive survey instrument that can aid national-scale travel survey.

5. Solutions: Designing a survey instrument for travel needs and barriers assessment

In this section, we review work that has contributed to systematic travel problem measurement. The reviewed studies will help create robust survey instrument for capturing the nuances of transportation challenges across different communities, ensuring that the voices of marginalized populations are represented in broader transportation planning efforts.

Murphy et al. (2022) introduce the Transportation Security Index (TSI), a 16-item survey tool designed to measure “transportation insecurity” across different demographic groups in the United States. Transportation insecurity is defined as the inability to regularly travel safely and in a timely manner. The study uses a nationally representative sample to document the prevalence of transportation insecurity, finding that one in four adults experience some form of transportation insecurity, with the highest rates observed among low-income, non-car owners, urban residents, and non-White populations. The TSI asks respondents how often they face challenges like being late, long waiting times, skipping trips, rescheduling trips, feeling stuck at home, or experiencing emotional distress, i.e., feeling left out, bad, embarrassed, worry about inconveniencing others, due to transportation issues, using a 3-point scale: ranging from never, sometimes, and often. These questions capture both material hardships (e.g., skipping trips,

waiting long times for transport) and subjective experiences (e.g., feeling embarrassed or worried about transport).

The “Freedom of Mobility Survey” by Singer and Martens (2023) is similar to the TSI. Like Murphy et al. (2022)’s 16-item TSI index, they directly asked individuals 12 different questions about their travel experiences and problems. The survey tool assesses three main categories of travel problems: 1) difficulties encountered while traveling, 2) reliance on others for travel, and 3) forgoing trips due to inadequate transportation. The questions are framed broadly, without asking for specific information about transport modes or destinations.

The Ease of Travel (EoT) study by De Vos (2024) introduces a person-based framework for assessing how easily people can travel. The study uses data from the 2023 University College London (UCL) travel survey, involving 2,593 students and staff, to test the EoT framework. Respondents were asked to evaluate 12 EoT-related statements on a five-point Likert scale, measuring aspects of motivation, skills, options, and quality of travel. Unlike the approach taken by Singer & Martens, which centered on survey questions targeting specific issues such as excessive travel time, discomfort, and physical or financial burdens, De Vos shifts the emphasis towards understanding the factors that facilitate or hinder smooth travel experiences. Furthermore, De Vos’ framework extends beyond these practical considerations to also consider personal motivations and skills, elements that Singer and Martens (2023) do not specifically address.

In these three studies, the response options are carefully designed to gauge the frequency of these issues rather than forcing a binary yes/no answer. This allows the survey to differentiate between respondents who experience occasional travel difficulties and those for whom travel problems are a regular occurrence, making the data more nuanced and useful for research. As demonstrated in Chapter 2, the NHTS collects data that overlap with some aspects of the TSI index, Freedom of Mobility Survey, and Ease of Travel (EoT) but also shows gaps in coverage. Below, we discuss ways that the NHTS could be enhanced through the inclusion of additional questions.

5.1 Perceived accessibility

NHTS can facilitate questions regarding individuals' lived experiences and perceptions regarding accessibility, emphasizing participatory processes to gain a deeper understanding of the barriers people face in their daily mobility. Studies based on NHTS data often focus on mobility rather than accessibility due to data limitations, but higher mobility does not necessarily equate with higher accessibility. For example, a person living in a suburban area with good mobility (e.g., owning a car and having no physical limitations) may still experience poor accessibility if essential services such as grocery stores, healthcare facilities, and schools are far from their residential location. Studies often tabulate a higher level of mobility for individuals or groups/households with greater travel distances or higher numbers of person trips. While shorter trip durations and lengths and lower numbers of person trips may not inherently indicate a problem, it prompts the critical question of whether this population is sacrificing access to potentially more suitable activity locations. Some people might travel less due to habit or personal choice, while others might be constrained by the external resources available to them. The lower frequency of trips among certain population groups, e.g., older adults, might be

interpreted as a focus on local and essential activities, aligning with expectations related to retirement and potential health considerations. The transportation equity literature increasingly focus on accessibility sufficiency, which focuses on ensuring that individuals have access to a minimum threshold of accessibility necessary to participate fully in society (Karner et al., 2023; Martens et al., 2022; Pot et al., 2024).

5.2 Latent transport demand

Another limitation that needs to be addressed in NHTS series is expanding more on unmet travel needs or latent transport demand related questions. Unrealized travel has been discussed in the literature with overlapping concepts like suppressed travel, forgone trips, and unmet need (Palm et al., 2024). Although the 2022 NHTS incorporated questions that covered reasons for fewer trips or not making any trips, more recent questions do not allow respondents an opportunity to explain the consequence of suppressed travel. Additionally, questions can explore whether certain trips were made from necessity rather than desire, such as trips to distant grocery stores due to the absence of local options. These questions will help to identify the extent to which individuals' mobility needs are being constrained and the impact of these constraints on their daily lives.

5.3 Travel motivation, skills, feelings, and satisfaction

Adding questions about skills, motivation, feelings, and attitudes to the NHTS can significantly enrich our understanding of the psychological and behavioral aspects of travel. These additions will offer insights into why people travel the way they do, beyond mere logistical needs, capturing personal and societal barriers to effective transportation. Different types of destinations have varying impacts on daily life, and ultimate life satisfaction depends on what individuals prioritize and how they are exposed to the city's transport system. Research has shown that travel satisfaction is a critical component of overall life satisfaction. Insufficient mobility and accessibility significantly impact travel satisfaction, which in turn affects overall life satisfaction, and quality of life (Delbosc et al., 2020; Ettema et al., 2010; R. J. Lee & Sener, 2016; Morris, 2015). According to De Vos et al. (2013) the ease and comfort of travel experiences are closely linked to individuals' overall happiness and contentment with their daily lives. Similarly, Ettema et al. (2011) found that positive travel experiences contribute significantly to overall life satisfaction, highlighting the importance of seamless and enjoyable travel in enhancing quality of life. The ability to reach desired destinations efficiently and comfortably contributes to a sense of autonomy and control over one's environment, which is a fundamental aspect of quality of life. The lack of access to necessary services not only hinders daily functioning but also limits opportunities for social interaction and community involvement, further diminishing quality of life. The NHTS can incorporate questions on travel satisfaction by asking respondents about their overall satisfaction with their daily travel experiences. Additionally, it can include specific questions that focus on satisfaction with different modes of transportation and the purpose of their travel, providing a more detailed understanding of how well travel needs are being met.

5.4 Sampling transport disadvantage groups adequately

The sample size in the NHTS may not be large enough to adequately analyze specific population groups. For example, information on immigrant nativity was not collected before 2001. The 2001 NHTS included questions about immigrant status, such as whether respondents were born in the U.S., their country of origin if not born in the U.S., and the year they entered the U.S. However,

these questions regarding country of origin or nativity were omitted in subsequent versions of the NHTS (2009, 2017, and 2022). The latest NHTS in 2022 no longer includes the question about year of entry. This omission forces researchers to use ethnic background as a proxy for immigration status (Delbosc & Shafi, 2023b). Further, race/ethnicity classifications based on the household's reference person could miss mixed-race dynamics.

To address the issue of limited sample sizes for specific population groups, the NHTS can incorporate several changes. First, implementing targeted oversampling of underrepresented groups such as immigrants, ethnic minorities, public transit users, and non-conforming gender groups will ensure a sufficient sample size for meaningful analysis of their travel behavior. Second, reintroducing and expanding questions regarding immigrant status, such as country of origin, year of entry into the U.S., and immigration status, will provide a more accurate picture of the travel behavior and needs of immigrant populations. Third, updating the race/ethnicity classification to allow for more detailed responses, including mixed-race categories, will help capture the diversity within households and provide a more nuanced understanding of travel patterns. Lastly, collaborating with community organizations and advocacy groups to reach and survey hard-to-reach populations will ensure higher response rates from diverse groups and help in designing culturally sensitive questions (Lubitow et al., 2021).

5.5 Built environment data

Another limitation related to NHTS data is that while spatial variables like urban area size, population and employment density, and the rural/urban status of respondents' home locations and trip ends are available, there is a lack of detailed neighborhood-built environment characteristics. This restriction hinders researchers from fully understanding the influence of local environments on travel behavior. To overcome this, researchers often rely on the confidential version of NHTS data to identify neighborhood characteristics of the sample respondents at the census tract and block group levels (Hu et al., 2021; Smart, 2015). These detailed characteristics are often measured in separate databases, such as the EPA's Smart Location Database. The NHTS can be updated to include more detailed questions about the immediate built environment of respondents. Questions can cover the presence and quality of sidewalks, bike lanes, public transit stops, green spaces etc.

6. Conclusion

The utilization of NHTS data in understanding the travel behaviors and needs of various population groups is invaluable. Yet, it is clear that these data alone are insufficient to comprehensively address the nuanced challenges of transport disadvantaged groups. The survey's focus on traditional metrics of mobility—such as trip frequency and mode of transport—while useful, tends to overlook the broader concept of accessibility and the qualitative aspects of travel experiences that significantly affect life satisfaction and social inclusion. To rectify this, future iterations of the NHTS should incorporate measures of travel satisfaction, perceived accessibility, and well-being impacts of travel experiences. These additions will allow for a deeper understanding of the barriers to mobility that go beyond physical and economic constraints, including those related to safety, comfort, and convenience.

Enhancing the NHTS will require a commitment to more inclusive, comprehensive data collection methods and a broader analytical perspective that considers both the quantitative and qualitative dimensions of travel. The next phase of work for the present project will involve refining the NHTS questionnaire to better capture the intricate aspects of modern travel behaviors, with a particular focus on underrepresented and transport-disadvantaged groups. This effort will aim to design a questionnaire that collects comprehensive mobility data while addressing identified gaps from previous surveys. To achieve this, we will incorporate qualitative metrics that assess perceived accessibility, travel satisfaction, and the emotional impacts of travel. Questions will be crafted to gauge respondents' satisfaction across various travel aspects including comfort, safety, convenience, and reliability, complemented by open-ended questions to gather personal narratives and deeper insights into travel choices.

To enhance representativeness, the survey instrument will strategically include targeted oversampling of historically underrepresented groups such as racial minorities, immigrants, people with disabilities, and low-income households. Collaboration with community organizations will help reach these populations effectively and ensure the questionnaire is culturally sensitive and accessible. Adaptations will also be made to reflect emerging transportation trends like ridesharing and e-scooters, with questions designed to assess usage frequency, choice reasons, and satisfaction levels. Pilot testing of the new questionnaire across diverse population segments will identify any clarity, sensitivity, or flow issues, with subsequent adjustments based on focus group feedback and expert reviews.

The desired impacts of the work include transportation policy, planning, decision-making, and investments that directly address the needs of populations that have historically been and/or are currently underserved by the system. Understanding and addressing these needs is a critically important for achieving transportation equity. We will communicate that that transportation solutions are not one-size-fits-all, but rather tailored to the unique challenges faced by disadvantaged groups. The project will ultimately reconfigure the transportation landscape, placing historically marginalized and underserved communities at the center of all aspects of transportation planning and decision making.

Overall, there is a growing need to augment traditional travel surveys with a qualitative exploration of individuals' experiences, incorporating a broader lens that considers not only the quantity of travel but also the quality, equity, and social implications of the journeys made. This expanded approach is vital for crafting inclusive transportation policies that cater to the diverse and often unmet needs of various transport disadvantaged groups.

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Appendix A: Variables from NHTS series related to Transport Equity

Label	Name	Responses	NHTS	Comments
Unmet Needs/Latent Demand				
<u>Fewer trips</u> in past 30 days	USAGE1	-1 = Valid skip -9 = Not ascertained 01 = Yes, took fewer trips 02 = No, did not take fewer trips	2022	First time asked in NHTS
Reason <u>fewer trips</u> - • More deliveries • Didn't feel safe • Didn't fee clean/healthy • Not reliable • Did not go where needed • Unaffordable • Health problems • No time • COVID-19 • Other	USAGE2_1 to USAGE2_10	-1 = Valid skip -9 = Not ascertained 01 = [Selected reasons] 02 = Not Selected	2022	First time asked in NHTS

Label	Name	Responses	NHTS	Comments
Reason for <u>not taking trips</u> on travel day or Reason for no trips	SAMEPLC	-1 = Valid skip -7 = I prefer not to answer -8 = Don't know -9 = Not ascertained 01 = Personally sick or quarantining 02 = Vacation or personal day 03 = Caretaking 04 = Disabled or home-bound 05 = Worked at home (for pay) 06 = Not scheduled to work 07 = Household chores/projects 08 = Bad weather 09 = Out of country 10 = No transportation available 12 = School holiday/teacher workday 14 = Hospitalized or otherwise confined 15 = School from home 16 = Reluctant to travel 97 = Something else, please specify	2022, 2017	2022 version added additional 5 responses
<u>Last trip</u> before travel day	LSTTRDAY17	-1 = Appropriate skip -7 = I prefer not to answer -8 = I don't know -9 = Not ascertained 01 = The day before 02 = A few days before 03 = A week before	2017	Unlike past two survey (2001,2009), this question asked for categorical response
Approximate <u>number of days</u> since last trip/Num days since last trip before travel day	LSTTRDAY	Number entry with options: Not Ascertained Don't Know Refused Appropriate Skip	2009, 2001	

Label	Name	Responses	NHTS	Comments
Would like to <u>get out more</u> often	MOROFTEN	-1 = Appropriate skip -7 = Refused -8 = Don't know -9 = Not ascertained 01 = Yes 02 = No	2009	
Date of last trip before travel day, Month	LASTRPM	-1 = Appropriate Skip -7 = Refused -8 = Don't Know -9 = Not Ascertained 01 = January 02 = February 03 = March 04 = April 05 = May 06 = June 07 = July 08 = August 09 = September 10 = October 11 = November 12 = December	2001	

Label	Name	Responses	NHTS	Comments
Date of last trip before travel day, Year	LASTRPYY	-1 = Appropriate Skip -7 = Refused -8 = Don't Know -9 = Not Ascertained 1978 = 1978 1985 = 1985 1990 = 1990 1993 = 1993 1994 = 1994 1995 = 1995 1996 = 1996 1997 = 1997 1998 = 1998 1999 = 1999 2000 = 2000 2001 = 2001 2002 = 2002	2001	
Affordability				
Travel is a financial burden	PLACE	-7 = I prefer not to answer -8 = I don't know -9 = Not ascertained 01 = Strongly agree 02 = Agree 03 = Neither Agree or Disagree 04 = Disagree 05 = Strongly disagree	2017	Didn't continue this in 2022
Bicycle/ Walk/Public transportation to reduce financial burden of travel	BIKE2SAVE/ WALK2SAVE/ PTRANS	-7 = I prefer not to answer -8 = I don't know -9 = Not ascertained 01 = Strongly agree 02 = Agree 03 = Neither Agree or Disagree 04 = Disagree 05 = Strongly disagree	2017	Didn't continue this in 2022

Label	Name	Responses	NHTS	Comments
Price of gasoline affects travel		-7 = I prefer not to answer -8 = I don't know -9 = Not ascertained 01 = Strongly agree 02 = Agree 03 = Neither Agree or Disagree 04 = Disagree 05 = Strongly disagree	2017	Didn't continue this in 2022
Paid for parking at any time during travel day	PARK	-1=Valid skip 1 = Yes 2 = No	2022	Introduced in 2022
Paid for parking on this trip	PARK2	-1 = Valid skip 0001 = Paid to park 9996 = N/A Paid for by employer 9997 = N/A Paid for by school 9998 = Paid but I don't remember the cost 9999 = I did not pay for parking at this location	2022	Introduced in 2022
Amount paid for parking	PARK2_PAMOUNT	-1= Skip Responses= .25 to 900	2022	Introduced in 2022
Periodicity of parking payment	PARK2_PAYTYPE	-1 = Appropriate skip 01 = One Time 02 = Per Week 03 = Per Month 05 = Per Semester 06 = Per Year	2022	Introduced in 2022
Employer pays for discounted transit pass	EMPPASS	-1=Valid skip 1 = Yes 2 = No	2022	
Discounted transit pass used in past 30 days	TRNPASS	-1=Valid skip 1 = Yes 2 = No	2022	
Perceived broad Transport Issues				
Most important transportation issue	ISSUE	-1 = Appropriate skip -7 = Refused -8 = Don't know -9 = Not ascertained	2009	Dropped in 2017, 2022

Label	Name	Responses	NHTS	Comments
		01 = Highway congestion 02 = Access to / availability of public transit 03 = Lack of walkways or sidewalks 04 = Price of travel 05 = Aggressive / distracted drivers 06 = Safety concerns		
Respondent's view on • Safety concerns • Highway congestion • Price of travel (fees, tolls and gas) • Aggressive/distracted drivers • Access or availability of public transit • Lack of walkways or sidewalks	DTACDT DTCONJ DTCOST DTRAGE DTRAN DTWALK	-1 = Appropriate skip -7 = Refused -8 = Don't know -9 = Not ascertained 01 = A little issue 02 = A moderate issue 03 = A big issue	2009	
Respondent's view on • Worrying about a traffic accident • Highway congestion • Distracted drivers • Drunk drivers • Price of gasoline • Lack of walkways or sidewalks • Rough pavement or potholes • Aggressive drivers on the road • Drivers speeding • Traffic or road congestion • Number of large trucks on road	DTACDT DTCONJ DTDISTR DTDRUNK DTGAS DTNOWALK DTPVPOT DTRRAGE DTSPEED DTTIEUP DTTRUCKS	-1 = Appropriate Skip -7 = Refused -8 = Don't Know -9 = Not Ascertained 1 = Not a problem 2 = A little problem 3 = Somewhat of a problem 4 = Very much of a problem 5 = A severe problem	2001	
Public Transit Reliability				
Time waiting for public transit/Derived length of wait for public transit - minutes	TRWAITTM	In 2001, N/A Don't Know Refused Skip 0-4 min 5-9 min 10-19 min 20-29 min	2001, 2009, 2017	Dropped in 2022

Label	Name	Responses	NHTS	Comments
		30-39 min 40-49 min 50+ min In 2009, -1 = Appropriate skip -9 = Not ascertained 0-302 = 0-302 In 2017, Not ascertained I don't know I prefer not to answer Appropriate skip 0-125		
COVID-19 Impact				
COVID impact on - online purchases for home delivery - use of public transit - travel to a physical school/class location - travel to a physical work location	COV1_OHD COV1_PT COV1_SCH COV1_WK	-1=Valid skip 01=Temporary 02=Permanent 03=Don't know	2022	Introduced in 2022
Changes temporary or permanent- - Home delivery - Public Transit Use	COV2_OHD COV2_PT	-1=Valid skip 01=Temporary 02=Permanent 03=Don't know	2022	Introduced in 2022
First mile-last mile connection				
Mode used to get to/from public transit? Bus/Rail/POV/Subway/Walk/Other	TRACC TREGR	-1 = Appropriate skip -9 = Not ascertained 01 = Yes 02 = No	2017	Dropped in 2022
1 st /mode used to get to public transit 1 st /mode used from public transit to destination	TRACC1 TREGR1	-1 = Appropriate skip -7 = Refused -8 = Don't know -9 = Not ascertained 01 = Car 02 = Van	2009, 2001	Different options than 2017 survey

Label	Name	Responses	NHTS	Comments
		03 = SUV 04 = Pickup truck 05 = Other truck 07 = Motorcycle 08 = Light electric vehicle (golf cart) 09 = Local public bus 10 = Commuter bus 11 = School bus 12 = Charter/tour bus 13 = City to city bus 14 = Shuttle bus 15 = Amtrak/intercity train 16 = Commuter train 17 = Subway/elevated train 18 = Streetcar/trolley 19 = Taxicab 20 = Ferry 21 = Airplane 22 = Bicycle 23 = Walk 24 = Special transit - people with disabilities		
Trip Time to Transit Station in Minutes Time to Destination from Transit in Minutes	TRACCTM/ TREGRTM	Not ascertained I don't know I prefer not to answer Appropriate skip 0-300	2017	Dropped in 2022
Derived time to get to public transit/How long to destination from transit	TRACCTM/ TREGRTM	-1 = Appropriate skip -9 = Not ascertained 0-440 = 0-440	2009	
Time to get to public transit/ How long to destination from transit- minutes	TRACCTM/ TREGRTM	N/A Don't Know Refused Skip 0-4 min	2001	

Label	Name	Responses	NHTS	Comments
		5-9 min 10-19 min 20-29 min 30-39 min 40-49 min 50+ min		
Modal Choice/Multi-modal				
Mode used in last 30 days- - bike - bike share - e-scooters - motorcycle - public transit - rideshare - taxi service - walk	LAST30_BKSHR/ ESCT/ MTRC/ PT/ RDSHR/ TAXI/ WALK	-1 = Valid skip -9 = Not ascertained 01 = Yes 02 = No	2022	E-scooters use is introduced for the first time.
Count of App Usage- - Bike Share - Carshare Program Usage - Rideshare	BIKESHARE/CARSHAR E/ RIDESHARE	Not ascertained I don't know I prefer not to answer Appropriate skip 0-99	2017	Introduced for the first time
Reason for Not Walking/Biking More: - Infrastructure - Safety	BIKE_DFR BIKE_GKP WALK_DEF WALK_GKQ	In 2017 options were -1 = Appropriate skip -9 = Not ascertained 01 = No nearby paths or trails 02 = No sidewalks or sidewalks are in poor condition 03 = No nearby parks 04 = 01 and 02 05 = 01 and 03 06 = 02 and 03 07 = 01, 02, and 03	2017	Dropped in 2022
Walk/Bike issue: - Violence/crime along route - Distance between home and school - Speed of traffic along route	In 2009 SCHCRIM; SCHDIST; SCHSPD; SCHTRAF; SCHWTHR	In 2009 options were -1 = Appropriate skip -7 = Refused -8 = Don't know	2009	

Label	Name	Responses	NHTS	Comments
- Amount of traffic along route - Poor weather or climate in the area	In 2017	-9 = Not ascertained 01 = Not an issue 02 = A little bit of an issue 03 = Somewhat of an issue 04 = Very much an issue 05 = A serious issue		
Medical device used: - Cane/Walking stick - Motorized Scooter or wheelchair - Devices to aid the blind or visually impaired - Walker or crutches - No medical device - Dog Assistance [dropped in 2022]	W_CHAIR W_SCCH W_VISIMP W_WKCR W_DOG None	1=Valid skip -7=I prefer not to answer -8=Don't know -9=Not ascertained	2022, 2017	2017 version had 9 options, that has been aggregated to 5 in 2022
Personal/Social-demographic Barriers to Mobility				
Due to condition or disability- - Limited driving daytime - Travel not affected - Used bus or subway less frequently - Prefer not to answer if travel is affected - Asked others for rides - Given up driving - Used rideshare/reduced fare taxi [didn't included in 2001] - Used special transportation services Reduced travel	CONDNIGH CONDNONE CONDPUB CONDRF CONDRIDE CONDRIVE CONDSHARE CONDSPEC CONDTRAV	-1 = Valid skip -9 = Not ascertained 01 = Selected 02 = Not Selected	2022, 2017, 2009, 2001	
Have a med condition making travel difficult	MEDOCOND	-1 = Appropriate Skip -7 = Refused -8 = Don't Know -9 = Not Ascertained 1 = Yes 2 = No	2001	
Length of time with medical condition	MEDCOND6	-1 = Valid skip -7 = I prefer not to answer -8 = Don't know	2022, 2017, 2009,	2001- 2017 response

Label	Name	Responses	NHTS	Comments
		-9 = Not ascertained 01 = 6 months or less 02 = More than 6 months 03 = Entire life In 2001 -1 = Appropriate skip -7 = Refused -8 = Don't know -9 = Not ascertained 01 = 0-5 months 02 = 6-11 months 03 = 1-4 years 04 = 5-9 years 05 = 10 years or more 06 = All his/her life	2001	categories were similar